



Semester DB Project: Phase II

Part I: Review Feedback & Revisions

Three groups provided feedback during presentations, overall positive. The two main critiques were:

- Consistency and integrity enforcement: reviewer noted that duplicate or inconsistent data could still get through in areas like department names, abbreviations, research group names, and person emails. They also suggested reviewing view logic to ensure all relationships behave as intended.
- Redundancy in progress tracking: 2 reviewers flagged that having both a `progress` integer and a `status` string on the task table creates redundancy and suggested simplifying

Addressing the feedback:

- For the progress/status concern, we added a CHECK constraint locking `status` to four valid values (Completed, In Progress, Not Started, Cancelled) and constraining `progress` between 0 to 100. The fields serve different purposes: `status` describes the current state and `progress` tracks exact completion percentage.
- For the integrity concern, we added a CHECK constraint on `person.email` to enforce valid format and added `status` and date ordering constraints to `project` and `task`. Project status is constrained to Active, Completed, Cancelled, or On Hold. We also created three database user roles (`rg\_reader`, `rg\_member`, `rg\_admin`) to enforce access levels at the database level rather than the application level.

Part II: Implementation Of Revisions

```
55 DROP TABLE IF EXISTS `experiment`;
56 /*!40101 SET @saved_cs_client      = @@character_set_client */;
57 /*!50503 SET character_set_client = utf8mb4 */;
58 CREATE TABLE `experiment` (
59   `experiment_id` int NOT NULL AUTO_INCREMENT,
60   `title` varchar(100) CHARACTER SET utf8mb3 COLLATE utf8mb3_unicode_ci NOT NULL,
61   `objective` text CHARACTER SET utf8mb3 COLLATE utf8mb3_unicode_ci,
62   `description` text CHARACTER SET utf8mb3 COLLATE utf8mb3_unicode_ci,
63   `start_date` date DEFAULT NULL,
64   `end_date` date DEFAULT NULL,
65   `status` varchar(50) CHARACTER SET utf8mb3 COLLATE utf8mb3_unicode_ci NOT NULL,
66   PRIMARY KEY (`experiment_id`),
67   CONSTRAINT `chk_experiment_dates` CHECK (((`end_date` is null) or (`start_date` is null) or (`end_date` > `start_date`))),
68   CONSTRAINT `chk_experiment_status` CHECK ((`status` in (_utf8mb3'Active',_utf8mb3'Completed',_utf8mb3'Cancelled',_utf8mb3'On Hold'))),
69   CONSTRAINT `chk_experiment_title` CHECK ((trim(`title`) <> _utf8mb4''))
70 ) ENGINE=InnoDB AUTO_INCREMENT=31 DEFAULT CHARSET=utf8mb3 COLLATE=utf8mb3_unicode_ci;
71 /*!40101 SET character_set_client = @saved_cs_client */;
72
```

For the experiment we made sure each date was valid. Another change we did was make sure the status is constrained to 4 states. Another is making sure the title is valid as well.

```

141 DROP TABLE IF EXISTS `literature`;
142 /*!40101 SET @saved_cs_client      = @@character_set_client */;
143 /*!50503 SET character_set_client = utf8mb4 */;
144 CREATE TABLE `literature` (
145   `lit_id` int NOT NULL AUTO_INCREMENT,
146   `title` varchar(255) CHARACTER SET utf8mb3 COLLATE utf8mb3_unicode_ci NOT NULL,
147   `year` year NOT NULL,
148   `journal` varchar(255) CHARACTER SET utf8mb3 COLLATE utf8mb3_unicode_ci DEFAULT NULL,
149   `doi` varchar(255) CHARACTER SET utf8mb3 COLLATE utf8mb3_unicode_ci DEFAULT NULL,
150   `url` varchar(255) CHARACTER SET utf8mb3 COLLATE utf8mb3_unicode_ci DEFAULT NULL,
151   `theory` text CHARACTER SET utf8mb3 COLLATE utf8mb3_unicode_ci,
152   PRIMARY KEY (`lit_id`),
153   CONSTRAINT `chk_literature_doi` CHECK (((`doi` is null) or (`doi` like _utf8mb3'10.%'))),
154   CONSTRAINT `chk_literature_title` CHECK ((trim(`title`) <> _utf8mb3'')),
155   CONSTRAINT `chk_literature_url` CHECK (((`url` is null) or (`url` like _utf8mb4'http%'))),
156   CONSTRAINT `chk_literature_year` CHECK ((`year` between 1900 and 2100))
157 ) ENGINE=InnoDB AUTO_INCREMENT=31 DEFAULT CHARSET=utf8mb3 COLLATE=utf8mb3_unicode_ci;
158 /*!40101 SET character_set_client = @saved_cs_client */;
159

```

Here we made it so that the literature table is constrained accordingly to make sure that it has valid data included. For example we first made sure the title exists and the url is in a valid format. Finally we made sure that the literature was released between 1900 and 2100 to make sure only relevant literature is included in the dataset.

```

190 DROP TABLE IF EXISTS `Person`;
191 /*!40101 SET @saved_cs_client      = @@character_set_client */;
192 /*!50503 SET character_set_client = utf8mb4 */;
193 CREATE TABLE `Person` (
194   `ID` int NOT NULL AUTO_INCREMENT,
195   `name` varchar(100) CHARACTER SET utf8mb3 COLLATE utf8mb3_unicode_ci NOT NULL,
196   `email` varchar(100) CHARACTER SET utf8mb3 COLLATE utf8mb3_unicode_ci NOT NULL,
197   `role` varchar(100) CHARACTER SET utf8mb3 COLLATE utf8mb3_unicode_ci NOT NULL,
198   PRIMARY KEY (`ID`),
199   CONSTRAINT `chk_person_email` CHECK ((`email` like _utf8mb3'%@%.%')),
200   CONSTRAINT `chk_person_email_notempty` CHECK ((trim(`email`) <> _utf8mb4'')),
201   CONSTRAINT `chk_person_name` CHECK ((trim(`name`) <> _utf8mb3''))
202 ) ENGINE=InnoDB AUTO_INCREMENT=31 DEFAULT CHARSET=utf8mb3 COLLATE=utf8mb3_unicode_ci;
203 /*!40101 SET character_set_client = @saved_cs_client */;
204

```

This revision makes sure that the person table has a valid email and name. To do this we added a constraint to verify that each single user has an email address that is valid (contains the @ symbol) and also that the user has an email address.

```

275 DROP TABLE IF EXISTS `project`;
276 /*!40101 SET @saved_cs_client      = @@character_set_client */;
277 /*!50503 SET character_set_client = utf8mb4 */;
278 CREATE TABLE `project` (
279   `project_id` int NOT NULL AUTO_INCREMENT,
280   `title` varchar(100) CHARACTER SET utf8mb3 COLLATE utf8mb3_unicode_ci NOT NULL,
281   `description` text CHARACTER SET utf8mb3 COLLATE utf8mb3_unicode_ci,
282   `start_date` date DEFAULT NULL,
283   `end_date` date DEFAULT NULL,
284   `status` varchar(50) CHARACTER SET utf8mb3 COLLATE utf8mb3_unicode_ci NOT NULL,
285   PRIMARY KEY (`project_id`),
286   CONSTRAINT `chk_project_dates` CHECK (((`end_date` is null) or (`start_date` is null) or (`end_date` > `start_date`))),
287   CONSTRAINT `chk_project_status` CHECK ((`status` in (_utf8mb3'Active',_utf8mb3'Completed',_utf8mb3'Cancelled',_utf8mb3'On Hold'))),
288   CONSTRAINT `chk_project_title` CHECK ((trim(`title`) <> _utf8mb4'')),
289 ) ENGINE=InnoDB AUTO_INCREMENT=31 DEFAULT CHARSET=utf8mb3 COLLATE=utf8mb3_unicode_ci;
290 /*!40101 SET character_set_client = @saved_cs_client */;
291

```

For this revision we added constraints to the project table. These constraints aim to make sure that each project has a valid start and end data, and has a valid status. The project status must be in 1 of 4 states, Active, Completed, Cancelled, or On Hold. Furthermore we made sure that it has a title.

```

488 DROP TABLE IF EXISTS `task`;
489 /*!40101 SET @saved_cs_client      = @@character_set_client */;
490 /*!50503 SET character_set_client = utf8mb4 */;
491 CREATE TABLE `task` (
492   `task_id` int NOT NULL AUTO_INCREMENT,
493   `title` varchar(100) CHARACTER SET utf8mb3 COLLATE utf8mb3_unicode_ci NOT NULL,
494   `description` text CHARACTER SET utf8mb3 COLLATE utf8mb3_unicode_ci,
495   `status` varchar(50) CHARACTER SET utf8mb3 COLLATE utf8mb3_unicode_ci NOT NULL,
496   `due_date` date DEFAULT NULL,
497   `completed_date` date DEFAULT NULL,
498   `progress` int NOT NULL DEFAULT '0',
499   PRIMARY KEY (`task_id`),
500   CONSTRAINT `chk_task_status` CHECK ((`status` in (_utf8mb3'Completed',_utf8mb3'In Progress',_utf8mb3'Not Started',_utf8mb3'Cancelled'))),
501   CONSTRAINT `chk_task_title` CHECK ((trim(`title`) <> _utf8mb3'')),
502 ) ENGINE=InnoDB AUTO_INCREMENT=31 DEFAULT CHARSET=utf8mb3 COLLATE=utf8mb3_unicode_ci;
503 /*!40101 SET character_set_client = @saved_cs_client */;
504

```

This revision will edit the task table and do something similar that we did for the project table, constrain the task status to 4 states and verify that each task has a valid title.

Adjust Access Levels:

-- Read-only user (public/general access)

```
CREATE USER 'rg_reader'@'localhost' IDENTIFIED BY 'ReaderPass123!';
```

```
GRANT SELECT ON research_group_db.* TO 'rg_reader'@'localhost';
```

-- Standard member (can insert and update, no delete)

```
CREATE USER 'rg_member'@'localhost' IDENTIFIED BY 'MemberPass123!';
```

```
GRANT SELECT, INSERT, UPDATE ON research_group_db.* TO 'rg_member'@'localhost';
```

-- Admin only (full access including delete)

```
CREATE USER 'rg_admin'@'localhost' IDENTIFIED BY 'AdminPass123!';
```

```
GRANT ALL PRIVILEGES ON research_group_db.* TO 'rg_admin'@'localhost';
```

```
FLUSH PRIVILEGES;
```

`rg\_reader` is for public or general access with SELECT only. `rg\_member` covers standard users who can add and update data but cannot delete. `rg\_admin` has full privileges including delete. This ensures users can only perform actions appropriate to their role.

Server: localhost:3306 » Database: keelyfow_research_group_db » Table: Person

Showing rows 0 - 24 (30 total, Query took 0.0004 seconds.)

SELECT * FROM `Person`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> | Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	ID	name	email	role
☐ Edit ☐ Copy ☐ Delete	1	Alice Johnson	alice.johnson@uni.edu	Principal Investigator
☐ Edit ☐ Copy ☐ Delete	2	Bob Martinez	bob.martinez@uni.edu	Principal Investigator
☐ Edit ☐ Copy ☐ Delete	3	Carol Chen	carol.chen@uni.edu	Lead Data Scientist
☐ Edit ☐ Copy ☐ Delete	4	David Kim	david.kim@uni.edu	Senior Researcher
☐ Edit ☐ Copy ☐ Delete	5	Eva Patel	eva.patel@uni.edu	Principal Investigator
☐ Edit ☐ Copy ☐ Delete	6	Frank Müller	frank.muller@uni.edu	Researcher
☐ Edit ☐ Copy ☐ Delete	7	Grace Okafor	grace.okafor@uni.edu	Research Associate
☐ Edit ☐ Copy ☐ Delete	8	Henry Zhao	henry.zhao@uni.edu	Senior Researcher
☐ Edit ☐ Copy ☐ Delete	9	Isla Rivera	isla.rivera@uni.edu	Postdoctoral Fellow
☐ Edit ☐ Copy ☐ Delete	10	James Novak	james.novak@uni.edu	Principal Investigator
☐ Edit ☐ Copy ☐ Delete	11	Laura Schmidt	laura.schmidt@uni.edu	Principal Investigator
☐ Edit ☐ Copy ☐ Delete	12	Marcus Thompson	marcus.thompson@uni.edu	Senior Researcher
☐ Edit ☐ Copy ☐ Delete	13	Nadia Volkov	nadia.volkov@uni.edu	Lead Engineer
☐ Edit ☐ Copy ☐ Delete	14	Omar Hassan	omar.hassan@uni.edu	Principal Investigator
☐ Edit ☐ Copy ☐ Delete	15	Priya Nair	priya.nair@uni.edu	Principal Investigator
☐ Edit ☐ Copy ☐ Delete	16	Quinn Fitzgerald	quinn.fitzgerald@uni.edu	Senior Researcher
☐ Edit ☐ Copy ☐ Delete	17	Rosa Delgado	rosa.delgado@uni.edu	Lead Engineer
☐ Edit ☐ Copy ☐ Delete	18	Samuel Osei	samuel.osei@uni.edu	Research Associate
☐ Edit ☐ Copy ☐ Delete	19	Tanya Bergström	tanya.bergstrom@uni.edu	Postdoctoral Fellow
☐ Edit ☐ Copy ☐ Delete	20	Umar Farooq	umar.farooq@uni.edu	Principal Investigator
☐ Edit ☐ Copy ☐ Delete	21	Vera Kowalski	vera.kowalski@uni.edu	Senior Researcher
☐ Edit ☐ Copy ☐ Delete	22	William Tran	william.tran@uni.edu	Postdoctoral Fellow
☐ Edit ☐ Copy ☐ Delete	23	Xena Papadopoulos	xena.papadopoulos@uni.edu	Lead Engineer
☐ Edit ☐ Copy ☐ Delete	24	Yusuf Adeyemi	yusuf.adeyemi@uni.edu	Research Associate
☐ Edit ☐ Copy ☐ Delete	25	Zoe Lindqvist	zoe.lindqvist@uni.edu	Principal Investigator

The Browse tab shows all 30 people in the database, each with a properly formatted email. This confirms our email constraint is doing its job since every entry follows the correct format with an @ symbol. Any bad email would get rejected on insertion

phpMyAdmin

Server: localhost:3306 Database: keelylow_research_group_db Table: project

Showing rows 0 - 24 (30 total, Query took 0.0004 seconds.)

SELECT * FROM `project`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	project_id	title	description	start_date	end_date	status
☐ Edit Copy Delete	1	CRISPR Gene Editing Study	Investigating CRISPR-Cas9 efficacy in human cell l...	2023-01-15	2024-12-31	Active
☐ Edit Copy Delete	2	Quantum Error Correction	Developing algorithms to minimize quantum decohere...	2022-06-01	2025-05-31	Active
☐ Edit Copy Delete	3	AI Tumor Detection	CNN-based model for early-stage tumor detection in...	2023-03-01	2024-09-30	Completed
☐ Edit Copy Delete	4	Arctic Ice Melt Simulation	Modeling ice sheet dynamics under climate change s...	2021-09-01	2023-08-31	Completed
☐ Edit Copy Delete	5	Synaptic Plasticity Analysis	Studying long-term potentiation in hippocampal neu...	2023-07-01	NULL	Active
☐ Edit Copy Delete	6	Biodegradable Scaffold Design	Creating scaffolds for bone tissue engineering	2022-11-01	2024-10-31	Active
☐ Edit Copy Delete	7	Social Media Sentiment Mining	Large-scale NLP analysis of public opinion trends	2023-05-15	2024-05-14	Completed
☐ Edit Copy Delete	8	Terahertz Imaging System	Building a sub-millimeter wave imaging device	2024-01-01	2026-01-01	Active
☐ Edit Copy Delete	9	Amyloid Fibril Structure	Cryo-EM study of amyloid aggregation pathways	2023-02-01	2025-01-31	Active
☐ Edit Copy Delete	10	Drug-Receptor Binding Simulation	MD simulations for novel kinase inhibitors	2022-04-01	2024-03-31	Completed
☐ Edit Copy Delete	11	CAR-T Cell Optimization	Engineering chimeric antigen receptors for improve...	2023-04-01	2025-03-31	Active
☐ Edit Copy Delete	12	COVID-19 Variant Surveillance	Genomic epidemiology of emerging SARS-CoV-2 varian...	2022-01-01	2023-12-31	Completed
☐ Edit Copy Delete	13	Graphene Transistor Fabrication	Developing sub-10nm graphene field-effect transist...	2023-06-01	NULL	Active
☐ Edit Copy Delete	14	Exoplanet Atmosphere Analysis	Spectroscopic characterization of habitable zone e...	2022-08-01	2024-07-31	Active
☐ Edit Copy Delete	15	Working Memory and Aging	fMRI study of prefrontal cortex activity in aging ...	2023-09-01	2025-08-31	Active
☐ Edit Copy Delete	16	Deep Sea Carbon Flux Study	Measuring biological pump efficiency in abyssal zo...	2021-05-01	2023-04-30	Completed
☐ Edit Copy Delete	17	Laparoscopic Robot Arm v2	Second-generation autonomous laparoscopic surgical...	2023-10-01	2026-09-30	Active

The project table shows all 30 projects, each with a status of Active, Completed, or On Hold confirming our status constraint is working. You can also see every end date comes after the start date, which confirms the date ordering constraint is enforced.

phpMyAdmin

Server: localhost:3306 » Database: keelyflow_research_group_db » Table: task

Showing rows 0 - 24 (30 total, Query took 0.0004 seconds.)

SELECT * FROM `task`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> | Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	task_id	title	description	status	due_date	completed_date	progress
☐ Edit Copy Delete	1	Design sgRNA sequences	Design 5 guide RNAs targeting BRCA1 locus	Completed	2023-02-28	2023-02-25	100
☐ Edit Copy Delete	2	Set up dilution refrigerator	Install and cool down the dilution refrigerator	Completed	2022-07-15	2022-07-12	100
☐ Edit Copy Delete	3	Train CNN baseline model	Train ResNet-50 on initial labeled MRI set	Completed	2023-07-01	2023-06-28	100
☐ Edit Copy Delete	4	Write climate model report	Summarize findings from simulation runs	Completed	2023-06-30	2023-07-02	100
☐ Edit Copy Delete	5	Prepare neuron staining protocol	Optimize immunofluorescence staining for NMDA rece...	In Progress	2024-01-15	NULL	60
☐ Edit Copy Delete	6	3D print scaffold prototypes	Print 10 prototype scaffolds using PLA and HA blen...	Completed	2023-03-31	2023-03-29	100
☐ Edit Copy Delete	7	Build sentiment classifier	Fine-tune BERT model on labeled tweet dataset	Completed	2023-09-01	2023-08-30	100
☐ Edit Copy Delete	8	Align optical components	Align beam path for THz spectrometer setup	In Progress	2024-06-01	NULL	45
☐ Edit Copy Delete	9	Collect cryo-EM images	Image 200 grid squares at 300 kV	In Progress	2024-02-01	NULL	70
☐ Edit Copy Delete	10	Validate docking poses	Cross-check MD poses with experimental IC50 data	Completed	2022-09-30	2022-09-28	100
☐ Edit Copy Delete	11	Clone CAR construct into lentiviral vector	Gibson assembly of CD19-targeting CAR into pLenti ...	Completed	2023-04-30	2023-04-28	100
☐ Edit Copy Delete	12	Build variant phylogenetic tree	Maximum likelihood tree of 500 sequenced isolates ...	Completed	2022-06-30	2022-07-01	100
☐ Edit Copy Delete	13	Deposit graphene onto SiO2 substrate	Wet transfer of CVD graphene onto 300nm SiO2/Si wa...	Completed	2023-11-30	2023-11-25	100
☐ Edit Copy Delete	14	Cross-match JWST targets with Gaia catalog	Astrometric cross-matching of 40 planet-hosting st...	Completed	2022-10-31	2022-10-29	100

The task table shows all 30 tasks with status values of either Completed or In Progress, confirming the status constraint is working. The progress column shows 100 for completed tasks and partial values like 45, 60, and 70 for in progress ones, confirming the 0 to 100 constraint is enforced.

Query 1:

```
SELECT name, email, role FROM Person;
```

✓ Showing rows 0 - 24 (30 total, Query took 0.0003 seconds.)

`SELECT name, email, role FROM Person;`

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

1 > >> | ☐ Show all | Number of rows: 25 | Filter rows: Sort by key: None

Extra options

name	email	role
Alice Johnson	alice.johnson@uni.edu	Principal Investigator
Bob Martinez	bob.martinez@uni.edu	Principal Investigator
Carol Chen	carol.chen@uni.edu	Lead Data Scientist
David Kim	david.kim@uni.edu	Senior Researcher
Eva Patel	eva.patel@uni.edu	Principal Investigator
Frank Müller	frank.muller@uni.edu	Researcher
Grace Okafor	grace.okafor@uni.edu	Research Associate
Henry Zhao	henry.zhao@uni.edu	Senior Researcher
Isla Rivera	isla.rivera@uni.edu	Postdoctoral Fellow
James Novak	james.novak@uni.edu	Principal Investigator
Laura Schmidt	laura.schmidt@uni.edu	Principal Investigator
Marcus Thompson	marcus.thompson@uni.edu	Senior Researcher
Nadia Volkov	nadia.volkov@uni.edu	Lead Engineer
Omar Hassan	omar.hassan@uni.edu	Principal Investigator
Priya Nair	priya.nair@uni.edu	Principal Investigator
Quinn Fitzgerald	quinn.fitzgerald@uni.edu	Senior Researcher
Rosa Delgado	rosa.delgado@uni.edu	Lead Engineer
Samuel Osei	samuel.osei@uni.edu	Research Associate
Tanya Bergström	tanya.bergstrom@uni.edu	Postdoctoral Fellow
Umar Farooq	umar.farooq@uni.edu	Principal Investigator
Vera Kowalski	vera.kowalski@uni.edu	Senior Researcher
William Tran	william.tran@uni.edu	Postdoctoral Fellow
Xena Papadopoulos	xena.papadopoulos@uni.edu	Lead Engineer
Yusuf Adeyemi	yusuf.adeyemi@uni.edu	Research Associate
Zoe Lindqvist	zoe.lindqvist@uni.edu	Principal Investigator

This query pulls all 30 people in the database with their name, email, and role. Every email follows the valid format we enforced with `chk\_person\_email` since each one has an @ symbol and a proper domain, confirming the constraint is working.

Query 2:

```
SELECT p.name, pr.title, pr.status, pm.role
FROM Person p
JOIN project_member pm ON p.ID = pm.person_id
```

JOIN project pr ON pm.project_id = pr.project_id;

✓ Showing rows 0 - 24 (30 total, Query took 0.0004 seconds.)

```
SELECT p.name, pr.title, pr.status, pm.role FROM Person p JOIN project_member pm ON p.ID = pm.person_id JOIN project pr ON pm.project_id = pr.project_id;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

1 > >> | ☐ Show all | Number of rows: 25 | Filter rows: Sort by key: None

Extra options

name	title	status	role
Alice Johnson	CRISPR Gene Editing Study	Active	Project Lead
Bob Martinez	Quantum Error Correction	Active	Project Lead
Carol Chen	AI Tumor Detection	Completed	ML Engineer
David Kim	Arctic Ice Melt Simulation	Completed	Climate Modeler
Eva Patel	Synaptic Plasticity Analysis	Active	Neuroscientist
Frank Müller	Biodegradable Scaffold Design	Active	Biomaterials Engineer
Grace Okafor	Social Media Sentiment Mining	Completed	NLP Analyst
Henry Zhao	Terahertz Imaging System	Active	Optical Engineer
Isla Rivera	Amyloid Fibril Structure	Active	Structural Biologist
James Novak	Drug-Receptor Binding Simulation	Completed	Computational Chemist
Laura Schmidt	CAR-T Cell Optimization	Active	Project Lead
Marcus Thompson	COVID-19 Variant Surveillance	Completed	Genomic Epidemiologist
Nadia Volkov	Graphene Transistor Fabrication	Active	Nanofabrication Engineer
Omar Hassan	Exoplanet Atmosphere Analysis	Active	Observational Astronomer
Priya Nair	Working Memory and Aging	Active	Cognitive Neuroscientist
Quinn Fitzgerald	Deep Sea Carbon Flux Study	Completed	Marine Chemist
Rosa Delgado	Laparoscopic Robot Arm v2	Active	Robotics Engineer
Samuel Osei	Human Migration Genomics	Completed	Bioinformatician
Tanya Bergström	Urban Air Quality Monitoring	Active	Atmospheric Scientist
Umar Farooq	Hydrogel Viscoelasticity Study	Active	Rheologist
Vera Kowalski	MRSA Resistance Mechanism Mapping	Active	Microbiologist
William Tran	Hippocampal Place Cell Simulation	Active	Computational Modeler
Xena Papadopoulos	Droplet Microfluidics for Blood Typing	Active	Microfluidics Engineer
Yusuf Adeyemi	Migratory Bird Tracking Study	Completed	Field Ecologist
Zoe Lindqvist	Rare Earth Isotope Profiling	Active	Geochemist

This JOIN connects Person, project_member, and project to show each researcher, their role, and the project they're assigned to. The status column shows only Active or Completed across all 30 records, which reflects the chk_project_status constraint we added limiting status to four valid values.

Query 3:

```
SELECT status, COUNT(*) AS total_projects  
FROM project
```

GROUP BY status;

[Show query box](#)

 Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available. [?](#)

 Showing rows 0 - 1 (2 total, Query took 0.0002 seconds.)

```
SELECT status, COUNT(*) AS total_projects FROM project GROUP BY status;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25  Filter rows:

[Extra options](#)

status	total_projects
Active	21
Completed	9

☐ Show all | Number of rows: 25  Filter rows:

[Query results operations](#)

This groups all projects by status and counts how many fall into each category. 21 are Active and 9 are Completed with no other status values present, which is consistent with the status constraint defined on the project table